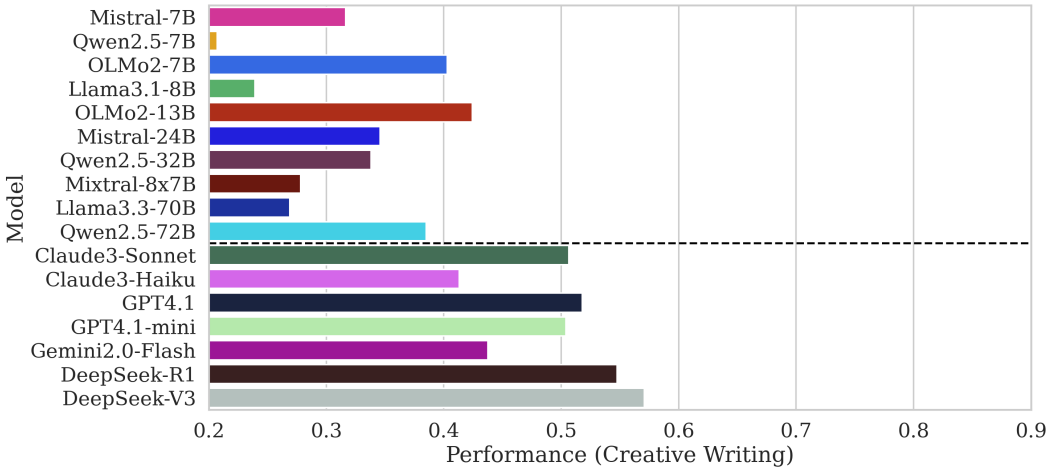

Figure 13: Performance on creative writing tasks

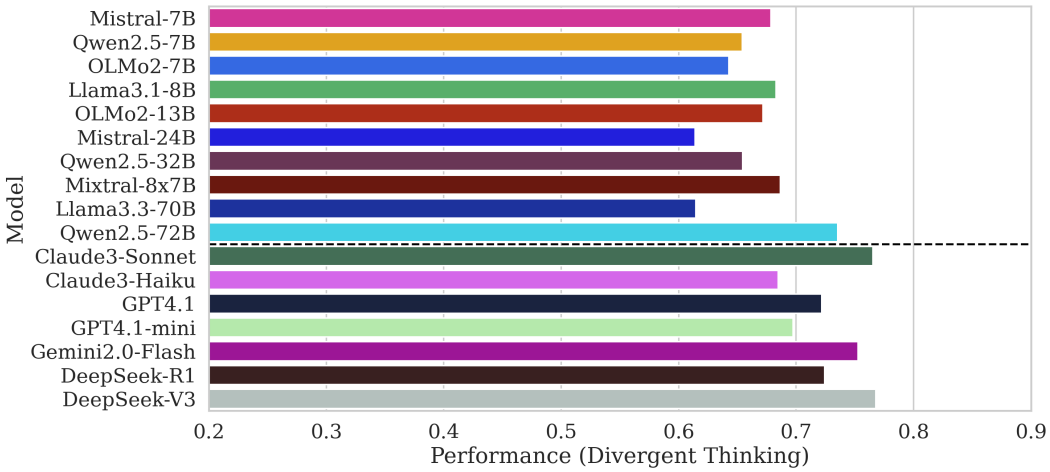

Figure 14: Performance on divergent thinking tasks

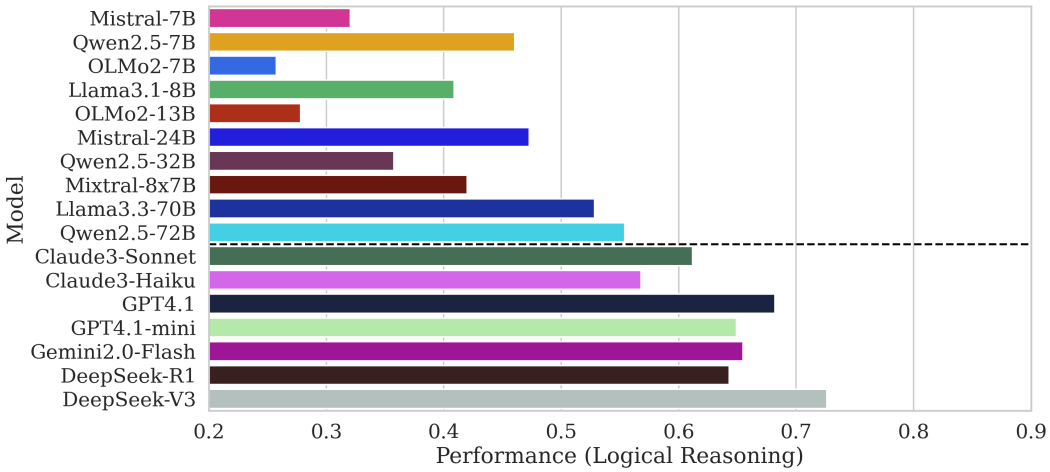

Figure 15: Performance on logical reasoning dimension

F Task Details

F.1 Torrance Test of Creative Writing (TTCW)

F.1.1 Dataset

The dataset consists of 12 New Yorker Stories' plots, i.e., GPT-4 generated summary of the original story [3].

F.1.2 Example

Plot

A woman experiences a disorienting night in a maternity ward where she encounters other similarly disoriented new mothers, leading to an uncanny mix-up where she leaves the hospital with a baby that she realizes is not her own, yet accepts the situation with an inexplicable sense of happiness.

Inference Prompt

Write a New Yorker-style story given the plot below. Make sure it is at least {word_count} words. Directly start with the story, do not say things like "Here's the story [...]" Plot: {plot} Story:

F.1.3 Experiment Configurations

- Temperature: 0.75
- Max Token: 4096
- Top-p: 1

F.1.4 Evaluation Metrics

As mentioned in Appendix D, we use a subset of questions from the original paper where experts achieved at least moderate agreement (Fleiss Kappa greater than 0.04) and our few-shot LLM-Judge passed the Alternative Annotator Test [9]. Since each evaluation question is binary for each generated story, we calculate the proportion of generated stories that pass each question as the final evaluation metric (e.g., if 3 out of 12 stories pass the "Narrative Ending (Fluency)" question, then the "Narrative Ending (Fluency)" metric is 0.25).

We use two-shot examples (one positive and one negative) in the evaluation prompt, as previous work shows adding few-shot examples improves human-LLM alignments [38].

[3] https://github.com/salesforce/creativity_eval

Evaluation Prompt

You are given a creative short story. Read it carefully. You are then given some background about specific aspects of creative writing, a binary (Yes/No) question, and sample stories with expert-annotated answers to the same question. Your objective is to use the background information and sample stories to answer the question about the story. Provide your answer in the format of "**Answer**: [Yes/No]". You can optionally then provide a short explanation for your answer.
==========
Question:
{full_prompt}

Examples:
==========
Story: {story}
Answer: {answer}
Explanations: {exp}
==========
Story: {story}
Answer: {answer}
Explanations: {exp}

==========
Story: {story}
Based on the question and examples above, answer the question (Provide your answer in the format of "**Answer**: [Yes/No]". You can optionally then provide a short explanation for your answer). Make sure you are extra harsh on the decision (most answers should be negative).
Answer:

F.1.5 Model Performances

Model	Narrative Ending (Fluency)	Perspective Voice (Flexibility)	Thought (Originality)	Form Structure (Originality)	Theme Content (Originality)
Mistral-7B	0.17	0.00	0.08	0.00	0.00
Qwen2.5-7B	0.00	0.00	0.17	0.00	0.00
OLMo2-7B	**0.67**	**0.17**	0.25	**0.08**	0.08
Llama3.1-8B	0.00	0.00	0.08	0.00	0.00
OLMo2-13B	0.50	0.00	**0.33**	**0.08**	**0.25**
Mistral-24B	0.08	0.00	0.25	0.00	**0.08**
Qwen2.5-32B	0.08	0.00	0.17	0.00	0.00
Mixtral-8x7B	0.08	0.00	0.08	0.00	0.00
Llama3.3-70B	0.00	**0.08**	0.33	0.00	0.00
Qwen2.5-72B	**0.50**	0.00	**0.50**	**0.17**	**0.08**
Claude3-Sonnet	0.75	**0.33**	0.58	0.42	**0.33**
Claude3-Haiku	0.50	0.00	0.25	0.08	0.00
GPT-4.1	0.75	0.25	**0.67**	0.50	**0.17**
GPT-4.1-mini	0.67	0.25	**0.83**	0.50	0.08
Gemini2.0-Flash	0.83	0.17	0.42	0.17	0.08
DeepSeek-R1	**0.92**	**0.33**	0.50	**0.58**	**0.17**
DeepSeek-V3	**1.00**	0.17	0.50	**0.50**	0.08

Table 7: Model performance on TTCW.